Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-7 | Summer-2025

Course Code	:	2101CS701	**Date**	:	21-04-2025
Course Name	:	Compiler Design	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Discuss types of translators. **4**

(B) Compare compiler v/s interpreter. **3**

OR

Compare front end and back end.

(C) Explain Phases of compiler. **7**

OR

Explain cousins of compiler.

Q.2 **(A)** Discuss input buffering methods. **4**

(B) Explain token, pattern, and lexemes. **3**

OR

Explain types of finite automata.

(C) Design NFA using Thompson's notation and then convert it into DFA for given regular expression: **7**
(aIb)*abb

OR

Design DFA without constructing NFA for given regular expression:
(aIb)*abb#

Q.3 **(A)** Explain synthesized attributes and inherited attributes. **4**

(B) Explain Recursive Decent Parsing method. **3**

OR

Discuss backtracking with suitable example.

(C) Design LL(1) parsing table for the following Grammar: **7**
E → E+T | T
T → T*F | F
F → (E) | a

OR

Design CLR parsing table for the following grammar:

S→AA

A→ aA | b

Q.4	**(A)**	Express following statement in form of DAG. a=b*-c+b*-c	**4**
	(B)	Explain activation record.	**3**

OR

Explain activation tree.

	(C)	Explain quadruple, triple, and indirect triple with example.	**7**

OR

Discuss source language issues.

Q.5	**(A)**	Explain flowgraph with example.	**4**
	(B)	Discuss loop optimization.	**3**

OR

Explain peephole optimization.

	(C)	Discuss issues in the design of a code generator.	**7**

OR

Discuss transformation on basic block.

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